

The World's Most Widely Adopted AI Developer Tool

GitHub + Company

Our Agenda

1

The Challenge
DevEx

2

The Opportunity
Accelerate workflows

3

The Platform
Trusted and proven AI

The DevEx difference.

Faster Time to Value

Grow revenue by improving DevEx:

- Shorter time to market
- Increased profitability

Source: Forrester, 2023

Improved Operational Efficiency

Drive business performance through increased:

- Productivity
- Efficiency
- Product quality

Source: Microsoft Research, DX, University of Victoria, 2023

Higher Talent Retention

Attract and retain top engineering talent:

- High team morale
- Motivating and rewarding work

Source: Gartner, 2023

- Intro

Expand Epics

shopping_cart_epics.md > # Shopping Cart System - Epic Level Work Items > ## Epic 3: Shopping Cart Operations

```
1  # Shopping Cart System - Epic Level Work Items
2  ## Epic 1: User Account Management
3  - User registration and account creation
4  - User authentication and authorization
5  - User profile management
6  - Password recovery and reset functionality
7  ## Epic 2: Product Catalog Management
8  - Product listing and categorization
9  - Product search functionality
10 - Product details and specifications display
11 - Inventory management integration
12 ## Epic 3: Shopping Cart Operations
13 - Add/remove items to/from cart
14 - Update item quantities
15 - Save cart for later
16 - Cart synchronization across devices
17 ## Epic 4: Checkout Process
18 - Shipping address management
19 - Shipping method selection
20 - Order review and confirmation
21 - Order submission processing
22 ## Epic 5: Payment Processing
23 - Multiple payment method support
24 - Secure payment processing
25 - Payment verification
26 - Refund and cancellation handling
27 ## Epic 6: Order Management
28 - Order history and status tracking
29 - Order modification capabilities
30 - Return and exchange processing
31 - Reorder functionality
32 ## Epic 7: Notification System
33 - Order confirmation notifications
34 - Shipping and delivery updates
35 - Stock alerts and back-in-stock notifications
36 - Price drop and promotion alerts
37 ## Epic 8: Analytics and Reporting
38 - User behavior tracking
```



Edit with Copilot

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Copilot is powered by AI, so mistakes are possible. Review output carefully before use.

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 or type # to attach context

@ to chat with extensions

Type / to use commands



Add Context...



shopping_cart_epics.md Current file



Edit files in your workspace



Edit ▾

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- Ask Mode

Create User Stories

m - Epic Level Work Items >  ## Epic 1: User Account Management >  ### Features:

```
1 # Shopping Cart System - Epic Level Work Items
4 ## Epic 1: User Account Management
6 **Business Value:** Enables personalized shopping
  experiences and secure access to the platform...
7 **Estimate:** Large (5-8 weeks)
8
9 ### Features:
10 1.1. **User Registration and Account Creation** -
    Medium (3-4 weeks)
11 --- Description: Enable new users to register and
    create accounts with email or social media
    integration
12 --- Acceptance Criteria: Email verification,
    duplicate prevention, GDPR compliance
13 --- Dependencies: None
14
15 1.2. **User Authentication and Authorization** -
    Medium (3-4 weeks)
16 --- Description: Implement secure login
    functionality with multi-factor authentication
    options
17 --- Acceptance Criteria: Session management,
    role-based access control, audit logging
18 --- Dependencies: 1.1
19
20 1.3. **User Profile Management** - Small (2 weeks)
21 --- Description: Allow users to view and edit
    personal information, preferences, and account
    settings
22 --- Acceptance Criteria: Profile completeness
    indicators, privacy controls
23 --- Dependencies: 1.1, 1.2
24
25 1.4. **Password Recovery and Reset Functionality** -
    Small (1-2 weeks)
26 --- Description: Provide secure mechanisms for
    users to recover access to their accounts
27 --- Acceptance Criteria: Time-limited reset
```



Ask Copilot

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 or type # to attach context

@ to chat with extensions

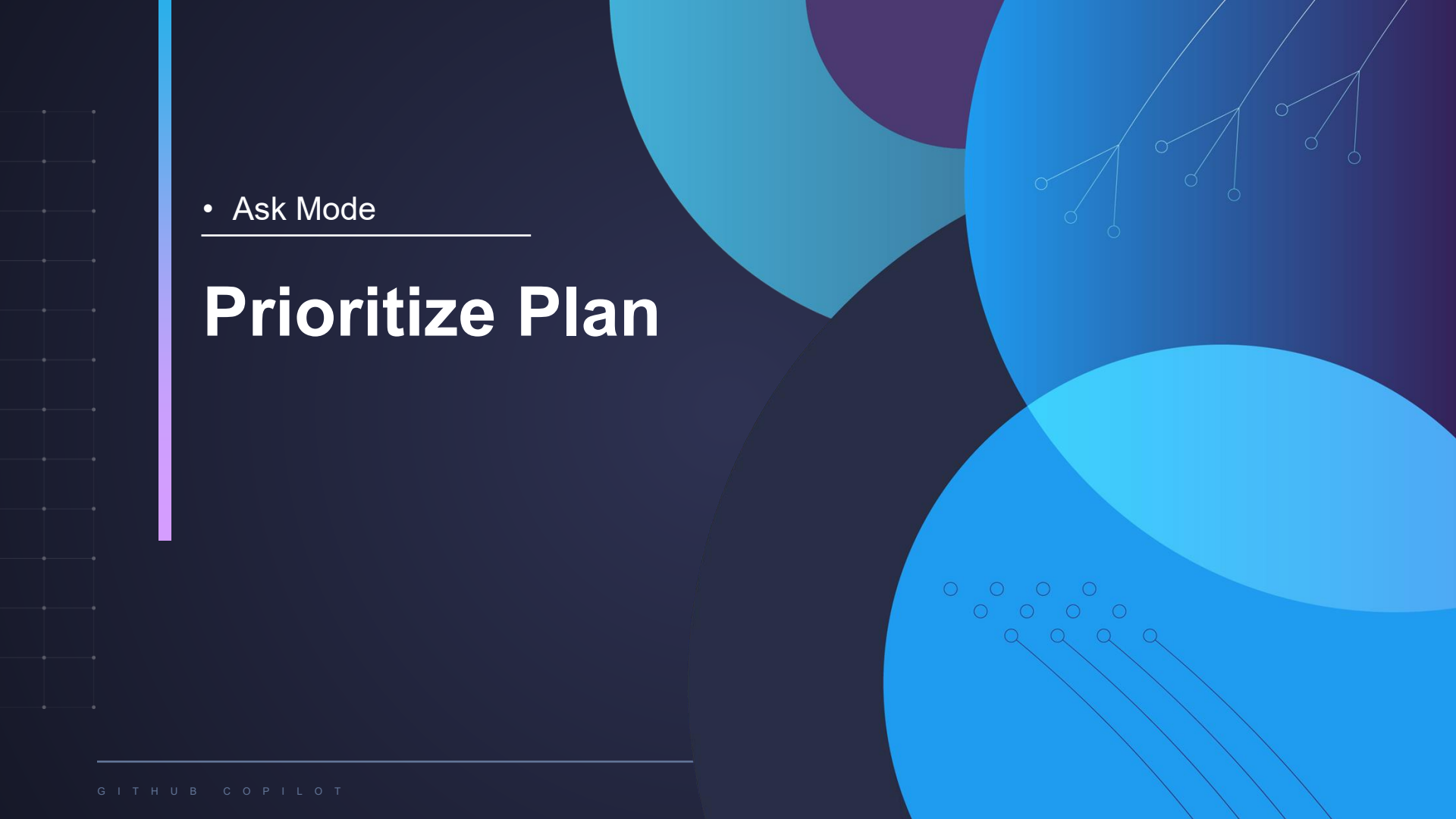
Type / to use commands

 Add Context... shopping_cart_epics.md:4-29 Current file 

Ask Copilot

I

@ Ask Claude 3.7 Sonnet Thinking 

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- Ask Mode
-

Prioritize Plan

m - Epic Level Work Items > [Epic] ## Epic 1: User Account Management > [Feature] ### Features:

```
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4 ## Epic 1: User Account Management
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23    --- Dependencies: 1.1, 1.2
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    - Small (1-2 weeks)
26    --- Description: Provide secure mechanisms for
    users to recover access to their accounts
27    --- Acceptance Criteria: Time-limited reset
    tokens, security questions, notification opt-in
```

- Option to request new token if expired
- Story Points: 5

1.4.3. As a user, I want to answer security questions as an alternative recovery method

- Acceptance Criteria:
 - Option to set up security questions during registration
 - Minimum of 2 questions required for recovery
 - Case-insensitive, but exact answer matching
 - Limit of 3 failed attempts before lockout
- Story Points: 8

1.4.4. As a user, I want to be notified when my password is changed

- Acceptance Criteria:
 - Immediate email notification of password change
 - Clear instructions if change wasn't authorized
 - Information about when and how password was changed
 - Contact support option within notification
- Story Points: 3

1.4.5. As a system administrator, I want to enforce password security

- Acceptance Criteria:
 - Password complexity requirements enforcement
 - Prevention of password reuse (last 5 passwords)
 - Secure password hashing implementation
 - Password expiration policy (optional for users)
- Story Points: 5

Each user story includes acceptance criteria and story point estimates to help with sprint planning and implementation tracking in accordance with Scrum/SAFe methodology.



Can you help with prioritization?

Add Context... shopping_cart_epics.md:4-29 Current file

Ask Copilot



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- Ask Mode

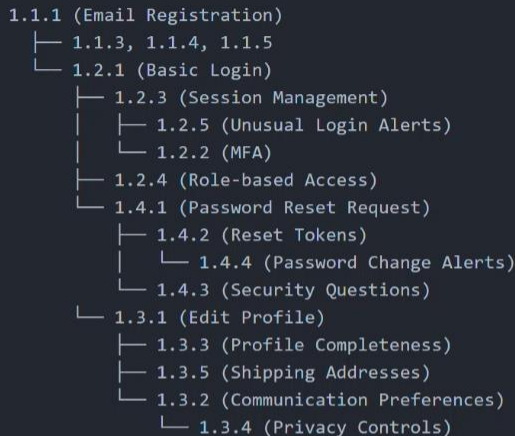
QA Test Cases

m - Epic Level Work Items > [Alt] ## Epic 1: User Account Management > [Esc] ### Features:

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4  ## Epic 1: User Account Management
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   - Small (1-2 weeks)
26   --- Description: Provide secure mechanisms for
   users to recover access to their accounts
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   tokens, security questions, notification on
```

- *Rationale:* Alternative recovery method
- *Dependencies:* 1.4.1, 1.4.2

Key Dependencies Visualization



This prioritization ensures:

1. Foundation is built first
2. Technical dependencies are respected
3. Core user flows are implemented early
4. Each phase delivers complete, testable functionality
5. Nice-to-have features come after essential ones



What about prioritizing by user impact?

📎 Add Context... 📄 shopping_cart_epics.md:4-29 Current file 🔗

Can you provide a sample QA test case for each user story? Don't generate code.

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@ 🗣

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- Ask Mode

QA Test Cases Summary

m - Epic Level Work Items > ## Epic 1: User Account Management > ### Features:

```
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        indicators, privacy controls
23     ...-Dependencies: 1.1, 1.2
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25 1.4. **Password Recovery and Reset Functionality** -
    Small (1-2 weeks)
26     ...-Description: Provide secure mechanisms for
        users to recover access to their accounts
27     ...-Acceptance Criteria: Time-limited reset
```

4. Test notification after password reset (forgotten flow)

5. Click "This wasn't me" link in notification

Expected Results:

1. Change notification email sent immediately
2. Email contains time and device information
3. Reset flow also triggers notification
4. "This wasn't me" option locks account pending verification
5. Security contact information present in email

Test Case 1.4.5: Password Complexity Requirements

User Story Reference: 1.4.5 - Enforce password security**Objective:** Verify password policy enforcement**Preconditions:** User is setting/changing password**Test Steps:**

1. Try password "password"
2. Try password "Password1"
3. Try password "P@ssword1"
4. Try current password (when changing)
5. Try password used previously
6. Try complex password "P@ssw0rd!2023"

Expected Results:

1. Simple password rejected with specific requirements
2. Password without special char rejected
3. Password meeting minimum requirements accepted
4. Current password rejected for changes
5. Previously used password rejected (last 5)
6. Complex new password accepted

Each test case includes detailed steps and expected results to verify the functionality meets acceptance criteria for the associated user story.



Can you summarize the test cases?



Add Context...



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- Ask Mode

Team Roles

opping Cart System - Epic Level Work Items > ## Epic 1: User Account Management > ### Features

```
1 # Shopping Cart System - Epic Level Work Items
2
3 ## Epic 1: User Account Management
4 authentication, and profile management...
5
6 **Business Value:** Enables personalized shopping
7 experiences and secure access to the platform...
8
9 **Estimate:** Large (5-8 weeks)
10
11 ### Features:
12 1.1. **User Registration and Account Creation** - Medium
13 (3-4 weeks)
14 --- Description: Enable new users to register and create
15 accounts with email or social media integration
16 --- Acceptance Criteria: Email verification, duplicate
17 prevention, GDPR compliance
18 --- Dependencies: None
19
20 1.2. **User Authentication and Authorization** - Medium
21 (3-4 weeks)
22 --- Description: Implement secure login functionality
23 with multi-factor authentication options
24 --- Acceptance Criteria: Session management, role-based
25 access control, audit logging
26 --- Dependencies: 1.1
27
28 1.3. **User Profile Management** - Small (2 weeks)
29 --- Description: Allow users to view and edit personal
30 information, preferences, and account settings
31 --- Acceptance Criteria: Profile completeness indicators,
32 privacy controls
33 --- Dependencies: 1.1, 1.2
34
35 1.4. **Password Recovery and Reset Functionality** - Small
36 (1-2 weeks)
37 --- Description: Provide secure mechanisms for users to
38 recover access to their accounts
39 --- Acceptance Criteria: Time-limited reset tokens,
40 security questions, notification on reset
41 --- Dependencies: 1.1, 1.2
```

- **Standard Login:** Tests basic authentication with valid credentials
- **Multi-Factor Authentication:** Validates MFA setup and verification workflow
- **Session Management:** Confirms proper timeout and renewal behavior
- **Role-Based Access:** Ensures different user roles have appropriate permissions
- **Suspicious Login Detection:** Verifies alerting for unusual login patterns

Feature 1.3: User Profile Management (5 test cases)

- **Profile Information Update:** Tests editing and validation of profile details
- **Notification Preferences:** Confirms customization of communication settings
- **Profile Completeness:** Validates progress indicators and suggestions
- **Privacy Settings:** Tests granular control over personal data
- **Address Management:** Verifies adding, editing, and deleting shipping addresses

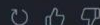
Feature 1.4: Password Recovery and Reset (5 test cases)

- **Password Reset Request:** Tests initiation of recovery workflow
- **Reset Token Validation:** Confirms token expiration and single-use behavior
- **Security Question Recovery:** Validates alternative recovery pathway
- **Password Change Notification:** Verifies alerts sent after password changes
- **Password Complexity:** Tests enforcement of security requirements

Key Coverage Areas

- **Security:** Authentication flows, MFA, session management, password policies
- **Data Protection:** GDPR compliance, privacy controls, consent management
- **Usability:** Profile management, notification preferences, address handling
- **Error Handling:** Invalid inputs, duplicate detection, rate limiting
- **Edge Cases:** Token expiration, multiple devices, concurrent sessions

Each test case includes detailed steps and expected results to validate the specific acceptance criteria outlined in the user stories, providing a comprehensive test suite for the User Account Management epic.



Can you prioritize these test cases?

📎 Add Context... 📄 shopping_cart_epics.md:4-29 Current file 🔗

Ask Copilot I



- Edit Mode

Create User Stories

shopping_cart_epics.md M X

Shopping Cart System - Epic Level Work Items > ## Epic 1: User Account Management

```

1  # Shopping Cart System - Epic Level Work Items
4  ## Epic 1: User Account Management
5  **Description:** Establish a robust user identity and
   access management system that supports account creation,
   authentication, and profile management.
6  **Business Value:** Enables personalized shopping
   experiences and secure access to the platform.
7  **Estimate:** Large (5-8 weeks)
8
9  ### Features:
10 1.1. **User Registration and Account Creation** - Medium
   (3-4 weeks)
11   ...- Description: Enable new users to register and create
   accounts with email or social media integration
12   ...- Acceptance Criteria: Email verification, duplicate
   prevention, GDPR compliance
13   ...- Dependencies: None
14
15 1.2. **User Authentication and Authorization** - Medium
   (3-4 weeks)
16   ...- Description: Implement secure login functionality
   with multi-factor authentication options
17   ...- Acceptance Criteria: Session management, role-based
   access control, audit logging
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21   ...- Description: Allow users to view and edit personal
   information, preferences, and account settings
22   ...- Acceptance Criteria: Profile completeness indicators,
   privacy controls
23   ...- Dependencies: 1.1, 1.2
24
25 1.4. **Password Recovery and Reset Functionality** - Small
   (1-2 weeks)
26   ...- Description: Provide secure mechanisms for users to
   recover access to their accounts
27   ...- Acceptance Criteria: Time-limited reset tokens,

```

CHAT



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or type # to attach context

to chat with extensions

Type / to use commands

Add Context... shopping_cart_epics.md:4-29 Current file

This is an Epic level Requirement with 4 features below. Can you help me create detailed features with user stories for each one listed. Number each item with the dot notation. Follow the Scrum / Scaled Agile methodology. Create a new file named for the Epic and add a link to the Epics file.



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- Edit Mode

QA Test Cases

stories > [M] ## Feature 1.4: Password Recovery and Reset Functionality > [M] ### User Stories

```
1 # Epic 1: User Account Management - Detailed User
  Scrum/Scaled Agile methodology.
5
6 ## Feature 1.1: User Registration and Account
  Creation
7
8 ### User Stories:
9
10 1.1.1. **Email Registration**
11 - **As a** new user,
12 - **I want to** create an account using my email
  address,
13 - **So that** I can access personalized shopping
  features.
14 - **Acceptance Criteria:**
15   - User can enter email, password, and required
  profile information
16   - Password strength requirements are clearly
  communicated
17   - Successful registration redirects to
  verification page
18   - Form includes validation for required fields
19   - Error messages are clear and actionable
20
21 1.1.2. **Social Media Registration**
22 - **As a** new user,
23 - **I want to** register using my social media
  accounts,
24 - **So that** I can create an account quickly
  without remembering another password.
25 - **Acceptance Criteria:**
26   - Support for major platforms (Google, Facebook,
  Apple, etc.)
27   - Required permissions clearly communicated to
  user
28   - Handling for when email from social account
  already exists in system
29   - Option to link additional social accounts later
```

This is an Epic level Requirement with 4 features below. Can you help me create detailed features with user stories for each one listed. Number each item with the dot notation. Follow the Scrum / Scaled Agile methodology. Create a new file named for the Epic and add a link to the Epics file.

[shopping_cart_epics.md:4-29](#)

GitHub Copilot

> Used 1 reference

I'll help you create detailed user stories for each feature under the User Account Management epic and create a dedicated file for this epic with a link from the main epics file.

Step-by-step solution:

1. Create a new markdown file called "user_account_management_epic.md"
2. Add detailed user stories for each of the 4 features, following Scrum/Agile methodology
3. Add a link to this new file from the main epics file

[shopping_cart_epics.md](#)

Add a link to the new detailed epic file at the end of Epic 1 section.

[shopping_cart_epics.md](#) +2 -0[user_account_management_epic.md](#)

Create a new file with detailed user stories for each feature.

[user_account_management_epic.md](#) +250 -1

2 files changed

[shopping_cart_epics.md](#)[user_account_management_epic.md](#)[Done](#)[📎 Add Context...](#)[user_account_management_epic.md](#) Current file 🔗

Edit files in **your workspace**



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- Edit Mode

QA Test Cases Summary

Detailed User Stories > **## Feature 1.1: User Registration and Account Creation** > **### User Stories:**

```
1  # Epic 1: User Account Management - Detailed User Stories
2  <!-- Generated by Jason Keicher and Copilot -->
3
4  This document contains detailed user stories for the User
5  Account Management epic, following the Scrum/Scaled Agile
6  methodology.
7
8  ## Feature 1.1: User Registration and Account Creation
9
10 ### User Stories:
11
12 1.1.1. **Email Registration**
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15 - **So that** I can access personalized shopping features.
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18   information
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30   - Support for major platforms (Google, Facebook, Apple,
31   etc.)
32   - Required permissions clearly communicated to user
33   - Handling for when email from social account already
34   exists in system
35   - Option to link additional social accounts later
36
37 1.1.3. **Email Verification**
38 - **As a** newly registered user,
```

GitHub Copilot

> Used 1 reference

I'll create a test case summary file that provides an overview of all QA test cases for the User Account Management epic.

Step-by-step solution:

1. Create a new summary file with an overview of all test cases
2. Include test case counts and key testing areas for each feature
3. Add links to the detailed test case files
4. Add a link back to the epic file

[test_cases_summary.md](#)

📌 test_cases_summary.md +71 -1

Now I'll update the original epic file to add a link to the test summary:

[user_account_management_epic.md](#)

📌 user_account_management_epic.md

🔄 👍 🗨

6 files changed

Done

📌 feature_1.1_test_cases.md
📌 feature_1.2_test_cases.md
📌 feature_1.3_test_cases.md
📌 shopping_cart_epics.md
📌 user_account_management_epic.md
📌 test_cases_summary.md

🔗 Add Context... 📌 user_account_management_epic.md Current file 🔗 📌 feature_1.1_test_cases.md X
📌 feature_1.2_test_cases.md X 📌 feature_1.3_test_cases.md X

Can you summarize the test cases and create a new summary file named for the Epic. Add the link back to the epic file.



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- Agent Mode

Create User Stories

- Agent Mode

Prioritize for Release Planning

- Agent Mode

QA Test Cases

